Finite-depth Quasi-Phase Matching (QPM) and Converter Efficiency

Abhishek V. Karve
Department of Applied Physics, Stanford University
(Dated: October 15, 2026)

An ensemble of ^{85}Rb atoms couples to optical (780 nm) and microwave cavities. A 480/481 nm blue control connects optical and Rydberg excitations; 297 nm UV dressing enables the effective microwave coupling from the prepared atomic state, completing the conversion loop [2]. A separate 1188 nm lattice confines the atoms and provides the periodic density for quasi-phase matching (QPM).

Finite site width reduces the coherent overlap Q between the two atomic conversion paths. This note derives the resulting efficiency, including cavity input and output coupling, and gives a numerical example. The derivation is self-contained; the companion Mathematica notebook implements and checks it, with a notation guide at the end [4].

In the level scheme of Ref. [2], g, e, r, f denote $5S_{1/2}$, $5P_{3/2}$, $36S_{1/2}$, and $35P_{1/2}$, respectively. UV dressing mixes $|g\rangle$ and $|f\rangle$ into the prepared reference state. Eliminating the unused UV-dressed state leaves the effective microwave matrix element g_m used below. For optical-to-microwave conversion, the loop absorbs an optical and a blue photon and emits a microwave and a UV photon. This sequence fixes the wavevector signs; the reversed sequence describes microwave-to-optical conversion.

I. WHAT THE FINITE-DEPTH QPM FACTOR MEANS

For atoms $j = 1, \ldots, N$, let u_j, v_j be the complex excitation amplitudes in the normalized optical/blue and UV/microwave collective profiles. They include relative coupling magnitudes and phases; overall strengths enter separately in section II. Define

$$Q = \left| \sum_j u_j^* v_j \right|, \qquad \sum_j |u_j|^2 = \sum_j |v_j|^2 = 1, \qquad 0 \le Q \le 1. \tag{1}$$

For matched, uniform envelopes, $Q = |\langle e^{i\Delta\mathbf{k}\cdot\mathbf{r}}\rangle|$, averaged over the normalized atomic spatial distribution, with $\Delta\mathbf{k} = \mathbf{k}_{780} + \mathbf{k}_{480} - \mathbf{k}_{\mathrm{UV}} - \mathbf{k}_{\mu}$. Wavevectors include propagation directions; reversing the loop reverses $\Delta\mathbf{k}$ without changing Q. Thus Q is an amplitude overlap and Q^2 the squared useful-mode projection. Conversion probability also depends on coherent exchange, atomic decay, and cavity ports. Neither Q nor Q^2 alone is a conversion probability.

A. Localization within one site.

An ideal retroreflected lattice has period $d = \lambda_{\mathrm{L}}/2$ and reciprocal wavevector $K = 2k_{\mathrm{L}}$, where $k_{\mathrm{L}} = 2\pi/\lambda_{\mathrm{L}}$. For displacement x from a site minimum, barrier height U, and atomic mass m, the harmonic approximation gives the recoil energy E_{R} and trap angular frequency ω_t [1]:

$$V(x) = U \sin^2(k_{\mathrm{L}} x) \simeq U k_{\mathrm{L}}^2 x^2, \quad E_{\mathrm{R}} = \frac{\hbar^2 k_{\mathrm{L}}^2}{2m}, \quad \omega_t = \frac{2\sqrt{U E_{\mathrm{R}}}}{\hbar}. \tag{2}$$

With reduced Planck constant $\hbar$, Boltzmann constant k_{B}, and temperature T, the thermal position variance and density Fourier harmonics are

$$\begin{aligned} \sigma_x^2 &= \frac{\hbar}{2m\omega_t} \coth\!\left(\frac{\hbar\omega_t}{2k_{\mathrm{B}}T}\right), \\ b_n &= \left|\left\langle e^{inKx}\right\rangle\right| = e^{-(nK)^2\sigma_x^2/2} = \exp\!\left[-n^2\sqrt{\frac{E_{\mathrm{R}}}{U}}\coth\!\left(\frac{\sqrt{U E_{\mathrm{R}}}}{k_{\mathrm{B}}T}\right)\right]. \end{aligned} \tag{3}$$

The integer n is a spatial harmonic order, not a vibrational band. At $T = 0$, coth $\to 1$, but zero-point motion keeps $b_n < 1$ for finite U and $n \neq 0$. Deeper trapping drives $b_n \to 1$ at fixed T. This approximation requires localization near the minimum; occupation near the barrier requires band wavefunctions.

B. From one axis to the total overlap.

Decompose position into the site centre $\mathbf{R}$ and the in-site displacement. Exact matching, $\Delta\mathbf{k} = \sum_i n_i \mathbf{K}_i$ with integer n_i, makes equivalent site centres contribute in phase. Finite in-site widths still reduce the overlap; for independent confined directions,

$$Q = \prod_i b_{n_i,i}; \qquad Q = b_{2,x} b_{2,y} \quad \text{for two second-order QPM axes.} \tag{4}$$

The last expression is the working geometry used below: co-propagating 780 and 480/481 nm fields, perpendicular UV, and suitable lattice axes. For an illustrative orthogonal-axis layout, take $\mathbf{K}_x = K\hat{\mathbf{x}}$, $\mathbf{K}_y = K\hat{\mathbf{y}}$, $\mathbf{k}_{780}, \mathbf{k}_{\text{blue}} \parallel +\hat{\mathbf{x}}$, and $\mathbf{k}_{\text{UV}} \parallel +\hat{\mathbf{y}}$. The target harmonic is then $(n_x, n_y) = (2, -2)$; $b_{-2} = b_2$. The component matching conditions are

$$\begin{aligned} k_{780} + k_{\text{blue}} - k_{\mu,x} &= 2K, \\ -k_{\text{UV}} - k_{\mu,y} &= -2K, \qquad k_{\mu,z} = 0. \end{aligned} \tag{5}$$

This is a *conditional geometry*, not fixed by the nominal wavelengths. For example, neglecting $\mathbf{k}_\mu$ and taking the labels literally gives $\delta k_x = -0.01017\,\text{rad}/\mu\text{m}$ for 480 nm and $-0.03738\,\text{rad}/\mu\text{m}$ for 481 nm, with 780 nm and 1188 nm held fixed. These correspond to about 1.02 and 3.74 radians across 100 μm. Actual wavelengths, beam angles, and reciprocal vectors must satisfy the vector condition; the numerical overlap below assumes that they do.

A residual mismatch $\delta\mathbf{k}$ produces phase cancellation across occupied site centres through $|\langle e^{i\delta\mathbf{k}\cdot\mathbf{R}}\rangle|$. For Gaussian site-centre widths L_i, this factor is $\exp[-\sum_i (\delta k_i L_i)^2/2]$, so $|\delta k_i| L_i \ll 1$ is a useful tolerance criterion. The in-site Fourier factor must also be evaluated at the actual mismatch. Exact site-centre matching aligns the phases from different wells; finite width still averages different phases within each well. Mode envelopes must be checked separately.

II. ANALYTICAL CONVERTER MODEL WITH THE ORTHOGONAL MODES

A. Problem definition and collective-mode construction.

The system contains an optical cavity mode $\hat{a}$, a microwave cavity mode $\hat{b}$, and an ensemble with optically excited and Rydberg excitations. The optical cavity couples to one collective atomic profile with strength g_o; the UV-dressed microwave interaction couples to another with strength g_m. The blue field connects the optical and Rydberg excitations with matrix element Ω. The profiles need not coincide: their normalized overlap is Q from Eq. (1).

We drive only the optical input and seek the coherent microwave output in the linear, weak-excitation regime. Efficiency is the ratio of output microwave photon flux to input optical photon flux; control fields carry the photon-energy difference. This is a photon-number efficiency, not an energy-efficiency ratio. All analytical couplings, detunings, and linewidths are angular frequencies; section III gives numerical values divided by 2π.

1. *Atomic variables and assumptions.*

Let $\hat{e}_j^\dagger$ and $\hat{r}_j^\dagger$ create optical and Rydberg excitations on atom j relative to the prepared UV-dressed reference state. The optical excitation is its coherence with $|e\rangle$; the Rydberg excitation is its coherence with $|r\rangle$, addressed by the blue and UV-dressed microwave paths. When signal excitations are few compared with the prepared atoms, depletion and saturation are negligible and these operators are approximately bosonic:

$$[\hat{e}_j, \hat{e}_\ell^\dagger] \simeq [\hat{r}_j, \hat{r}_\ell^\dagger] \simeq \delta_{j\ell}, \qquad [\hat{e}_j, \hat{r}_\ell^\dagger] \simeq 0. \tag{6}$$

Assume frozen positions, uniform blue-control magnitude, identical atomic detunings and independent decay rates. UV dressing is included in g_m and its profile. Absorbing the blue phase into $\hat{r}_j$ makes Ω real and uniform. It is the Hamiltonian matrix element below; convert any Rabi-frequency convention containing an extra factor of 1/2 before substitution:

$$\frac{\hat{H}_B}{\hbar} = -\Omega \sum_j \left(\hat{e}_j^\dagger \hat{r}_j + \hat{r}_j^\dagger \hat{e}_j\right). \tag{7}$$

2. The two coupling profiles.

In this phase convention, the optical cavity selects $\mathbf{u}$ and the UV-dressed microwave interaction selects $\mathbf{v}$, as in section I:

$$\mathbf{u}^\dagger\mathbf{u} = \mathbf{v}^\dagger\mathbf{v} = 1, \qquad \mathcal{Q} = \mathbf{u}^\dagger\mathbf{v} = \sum_j u_j^* v_j, \qquad Q = |\mathcal{Q}|. \tag{8}$$

Absorb the global phase of $\mathbf{v}$ into $\hat{b}$ to choose $\mathbf{u}^\dagger\mathbf{v} = Q \geq 0$. With $s = \sqrt{1-Q^2}$,

$$\boxed{\mathbf{v} = Q\mathbf{u} + s\mathbf{w},} \qquad \mathbf{w} = \frac{\mathbf{v} - Q\mathbf{u}}{s}, \qquad \mathbf{u}^\dagger\mathbf{w} = 0, \quad \mathbf{w}^\dagger\mathbf{w} = 1. \tag{9}$$

The formula for $\mathbf{w}$ applies for $Q < 1$. At $Q = 1$, any normalized orthogonal profile suffices because its microwave coupling vanishes ($s = 0$).

3. Collective operators and closure.

Define four collective operators (uppercase letters distinguish operators from their mean amplitudes):

$$\begin{aligned} \hat{P}^\dagger &= \sum_j u_j \hat{e}_j^\dagger, \quad \hat{R}^\dagger = \sum_j u_j \hat{r}_j^\dagger, \\ \hat{P}_\perp^\dagger &= \sum_j w_j \hat{e}_j^\dagger, \quad \hat{R}_\perp^\dagger = \sum_j w_j \hat{r}_j^\dagger. \end{aligned} \tag{10}$$

In the weak-excitation limit, orthonormality gives unit commutators within each mode; distinct modes commute. The coupling chain is

$$a \overset{g_o}{\longleftrightarrow} P \overset{\Omega}{\longleftrightarrow} R \overset{Qg_m}{\longleftrightarrow} b \overset{sg_m}{\longleftrightarrow} R_\perp \overset{\Omega}{\longleftrightarrow} P_\perp. \tag{11}$$

Here the labels denote modes, with hats suppressed. The orthogonal pair describes a different spatial pattern of excitation of the same atoms. It has no direct optical-cavity coupling but remains coupled to the microwave cavity and blue field. Orthogonality alone is not irreversible loss; atomic decay makes this channel dissipative. The two cavity coupling profiles span at most a two-dimensional atomic subspace, preserved by uniform control, detunings, and damping. Consequently, one additional optical/Rydberg pair suffices: the response closes on four atomic and two cavity modes. All other orthogonal atomic modes are initially unexcited and have zero mean drive.

4. Hamiltonian and conventions.

The collective matrix elements g_o, g_m include each profile's total coupling magnitude and are held fixed as Q varies. They are collective couplings, not single-atom matrix elements. The baseline parameter convention is [4]

$$g_o = \frac{G}{\sqrt{2}}, \qquad g_m = \frac{G_\mu}{\sqrt{2}}. \tag{12}$$

Thus G, G_μ rescale the same collective couplings. These factors reproduce the notebook convention; they are not an additional finite-depth correction. For experimental calibration, g_o, g_m should be identified with the measured collective matrix elements in Eq. (14). With zero optical-cavity detuning, the rotating-frame Hamiltonian fixes the signs of the optical atomic, Rydberg, and microwave detunings $\Delta_e, \Delta_r, \Delta_\mu$:

$$\frac{\hat{H}_0}{\hbar} = -\Delta_\mu \hat{b}^\dagger\hat{b} - \Delta_e(\hat{P}^\dagger\hat{P} + \hat{P}_\perp^\dagger\hat{P}_\perp) - \Delta_r(\hat{R}^\dagger\hat{R} + \hat{R}_\perp^\dagger\hat{R}_\perp), \tag{13}$$

$$\frac{\hat{H}_{\rm int}}{\hbar} = -\Big[g_o\hat{a}^\dagger\hat{P} + \Omega(\hat{P}^\dagger\hat{R} + \hat{P}_\perp^\dagger\hat{R}_\perp) + g_m\hat{b}^\dagger(Q\hat{R} + s\hat{R}_\perp) + \text{h.c.}\Big]. \tag{14}$$

Use $\dot{\hat{c}} = (i/\hbar)[\hat{H}, \hat{c}]$ and Fourier amplitudes $e^{+i\omega t}$, with spectral offset ω. For optical input and useful microwave output coupling rates $\kappa_{o,\rm ex}, \kappa_{m,\rm ex}$, the optical drive is

$$\frac{\hat{H}_{\rm d}}{\hbar} = -\sqrt{\kappa_{o,\rm ex}}\left[\hat{a}^\dagger a_{\rm in}(t) + a_{\rm in}^*(t)\hat{a}\right]. \tag{15}$$

Here $|a_{\rm in}|^2$ is incident photon flux. Write $a = \langle\hat{a}\rangle$, $b = \langle\hat{b}\rangle$, $p = \langle\hat{P}\rangle$, $r = \langle\hat{R}\rangle$, $p_\perp = \langle\hat{P}_\perp\rangle$, and $r_\perp = \langle\hat{R}_\perp\rangle$. With no microwave input, $b_{\rm out} = \sqrt{\kappa_{m,\rm ex}}\, b$ up to a port phase, and $S_{mo} = b_{\rm out}/a_{\rm in}$. Reservoir noise has zero mean; this response calculation does not characterize added noise.

B. Response functions and linewidth conventions.

Let κ_o, κ_m be the bare cavity energy decay rates, including all ports and internal cavity loss. Thus $\kappa_{o,\rm ex} \le \kappa_o$ and $\kappa_{m,\rm ex} \le \kappa_m$. The atomic linewidth parameters Γ, γ_r give optical and Rydberg amplitude decay rates $\Gamma/2, \gamma_r/2$, respectively.

Introduce an independent added microwave energy-loss rate $\gamma_{m,\rm add} \ge 0$ for any photon-loss reservoir not already included in κ_m or the explicit atomic dynamics. The baseline notebooks set $\gamma_{m,\rm add} = \gamma_r$ [4], but do not identify a microscopic reservoir for that choice. We retain this equality only for the legacy numerical benchmark. The linearized cavity equation in Ref. [3], Sec. B.3, instead has bare damping $\kappa_m/2$; it corresponds to $\gamma_{m,\rm add} = 0$ here. Atomic loss mediated by the retained Rydberg and optical modes is already accounted for by their response and must not be counted again as $\gamma_{m,\rm add}$. A physical use of nonzero $\gamma_{m,\rm add}$ therefore requires independent justification. Including damping, the inverse response functions are

$$\begin{aligned} D_a &= \kappa_o/2 + i\omega, & D_e &= \Gamma/2 + i(\omega - \Delta_e), \\ D_r &= \gamma_r/2 + i(\omega - \Delta_r), & D_b &= (\kappa_m + \gamma_{m,\rm add})/2 + i(\omega - \Delta_\mu). \end{aligned} \tag{16}$$

Real parts give amplitude damping; imaginary parts give detuning. Orthogonal-mode feedback is derived separately below.

C. Six coupled equations.

With these conventions,

$$\begin{aligned} D_a a - i g_o p &= i\sqrt{\kappa_{o,\rm ex}}\, a_{\rm in}, \\ D_e p - i g_o a - i\Omega r &= 0, \\ D_r r - i\Omega p - i g_m Q b &= 0, \\ D_b b - i g_m\big(Q r + \sqrt{1-Q^2}\, r_\perp\big) &= 0, \\ D_r r_\perp - i\Omega p_\perp - i g_m \sqrt{1-Q^2}\, b &= 0, \\ D_e p_\perp - i\Omega r_\perp &= 0. \end{aligned} \tag{17}$$

The microwave coupling splits into useful amplitude $Q g_m$ and orthogonal amplitude $g_m\sqrt{1-Q^2}$, retaining total strength g_m.

D. Eliminating the orthogonal pair.

Substitute $p_\perp = i\Omega r_\perp/D_e$ from the last equation into the preceding one. With $F = D_e D_r + \Omega^2$,

$$r_\perp = \frac{i g_m \sqrt{1-Q^2}\, D_e}{F}\, b, \qquad \boxed{\Sigma_\perp(\omega) = \frac{g_m^2 (1-Q^2) D_e}{F}}. \tag{18}$$

The microwave inverse response becomes $D_b + \Sigma_\perp$. The eliminated pair adds amplitude damping $\mathrm{Re}\,\Sigma_\perp$ (full linewidth $2\,\mathrm{Re}\,\Sigma_\perp$) and dispersion $\mathrm{Im}\,\Sigma_\perp$. Despite lacking direct optical-cavity coupling, it is microwave-driven, feeds back on that field, and dissipates through atomic decay. Replacing G_μ by QG_μ alone discards this feedback. Neither this damping nor the baseline extra damping belongs in the useful output-port rate.

E. Closed transfer function.

With $B_\perp = D_b + \Sigma_\perp$, eliminating the remaining atomic amplitudes gives

$$S_{mo} = \frac{\sqrt{\kappa_{o,\rm ex}\kappa_{m,\rm ex}}\, g_o g_m \Omega Q}{B_\perp(D_a F + g_o^2 D_r) + g_m^2 Q^2 (D_a D_e + g_o^2)}. \tag{19}$$

Set $Y = D_a F + g_o^2 D_r$, $Z = D_b F + g_m^2 D_e$. Substituting $\Sigma_\perp$ and $F = D_e D_r + \Omega^2$ gives, up to the port phase,

$$\boxed{S_{mo}(\omega) = \frac{\sqrt{\kappa_{o,\mathrm{ex}}\kappa_{m,\mathrm{ex}}}\, g_o g_m \Omega Q F}{YZ + Q^2 g_o^2 g_m^2 \Omega^2}, \qquad \eta(\omega) = |S_{mo}(\omega)|^2.} \tag{20}$$

Because Q enters both numerator and denominator, efficiency cannot generally be scaled by Q^2. At $Q = 1$ the orthogonal pair decouples, recovering the four-mode baseline; at $Q = 0$ coherent conversion vanishes.

F. Effective cavity coupling and matching.

Eliminating all four atomic amplitudes at once exposes the physical structure:

$$\begin{pmatrix} A & iJ \\ iJ & B \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} i\sqrt{\kappa_{o,\mathrm{ex}}}\, a_{\mathrm{in}} \\ 0 \end{pmatrix}, \qquad A = D_a + \frac{g_o^2 D_r}{F}, \quad B = D_b + \frac{g_m^2 D_e}{F}, \quad J = \frac{Q g_o g_m \Omega}{F}. \tag{21}$$

Here A and B are the fully dressed inverse cavity responses; B differs from the intermediate $B_\perp$. The response is $S_{mo} = \sqrt{\kappa_{o,\mathrm{ex}}\kappa_{m,\mathrm{ex}}}\, J/(AB + J^2)$, equivalent to Eq. (20). This frequency-domain elimination is exact within Eq. (17); no assumption of fast atomic relaxation is needed. Away from resonance, A, B, J are complex, frequency-dependent susceptibilities.

At fixed g_o, g_m, Ω and linewidths, the diagonal atomic loading of each cavity is independent of Q, while the cross-cavity coupling J is proportional to Q. The microwave mode still couples to its entire normalized atomic profile: its useful and orthogonal contributions to the diagonal loading add as $Q^2 + (1 - Q^2) = 1$. Reducing g_m to $Q g_m$ would reduce both the cross coupling and that total loading, explaining why it gives a different response.

On resonance, $A, B > 0$ and J are real. Define

$$\mathcal{C} = \frac{J^2}{AB}, \qquad \boxed{\eta(0) = \frac{\kappa_{o,\mathrm{ex}}}{2A}\frac{\kappa_{m,\mathrm{ex}}}{2B}\frac{4\mathcal{C}}{(1+\mathcal{C})^2}.} \tag{22}$$

The first two factors compare useful port escape with the dressed damping. The final factor describes coupling relative to damping and reaches unity at $\mathcal{C} = 1$: the two damped responses are matched. This condition maximizes conversion at fixed A, B, but need not eliminate reflection when internal losses are present. For $\mathcal{C} < 1$, weaker overlap moves the system farther from matching and lowers conversion. For $\mathcal{C} > 1$, reducing Q can instead improve conversion at the fixed resonant frequency by bringing $\mathcal{C}$ toward one. Thus Q is a mode overlap, not a universal multiplicative efficiency penalty; a loss budget also requires the reflected field.

III. ON-RESONANCE EFFICIENCY AND THE NUMERICAL VALUE OF Q

At $\omega = \Delta_e = \Delta_r = \Delta_\mu = 0$, define cooperativities comparing coherent coupling to damping:

$$C_o = \frac{2G^2}{\kappa_o \Gamma}, \qquad C_m = \frac{2G_\mu^2}{(\kappa_m + \gamma_{m,\mathrm{add}})\gamma_r}, \qquad C_B = \frac{4\Omega^2}{\Gamma\gamma_r}. \tag{23}$$

Let $\eta_o = \kappa_{o,\mathrm{ex}}/\kappa_o$ and $\eta_m = \kappa_{m,\mathrm{ex}}/\kappa_m$. Equation (20) becomes

$$\boxed{\eta(0) = \eta_o \eta_m \frac{\kappa_m}{\kappa_m + \gamma_{m,\mathrm{add}}} \frac{4Q^2 C_o C_m C_B (1 + C_B)^2}{\left[(1 + C_B + C_o)(1 + C_B + C_m) + Q^2 C_o C_m C_B\right]^2}.} \tag{24}$$

In these variables, the matching parameter is

$$\mathcal{C} = \frac{Q^2 C_o C_m C_B}{(1 + C_B + C_o)(1 + C_B + C_m)}. \tag{25}$$

Port fractions multiply the efficiency at fixed total linewidths. Changes to mirror or microwave-port coupling that alter linewidths require updated cooperativities. Propagation and detection transmissions further multiply this port-to-port result to give end-to-end efficiency.

A. Lattice example.

Take ^{85}Rb, $\lambda_{\mathrm{L}} = 1188\,\mathrm{nm}$, waist $w = 200\,\mu\mathrm{m}$, and 100% retroreflection. Adopt the specified calibration 3 W per beam per axis $\leftrightarrow U/k_{\mathrm{B}} = 15\,\mu\mathrm{K}$; no polarizability calculation is implied. With $E_{\mathrm{R}}/k_{\mathrm{B}} = 0.0799004\,\mu\mathrm{K}$, $T = 0$, and two equal, exactly matched second-order axes,

$$b_2 = e^{-4\sqrt{E_{\mathrm{R}}/U}} = 0.746816, \qquad \boxed{Q = b_2^2 = 0.557734}, \qquad Q^2 = 0.311067. \tag{26}$$

This is a harmonic ground-state benchmark. The relevant length and energy scales are

$$\begin{aligned} U/E_{\mathrm{R}} &= 187.7, & \sigma_x &= 36.12\,\mathrm{nm}, & 2K\sigma_x &= 0.7641, \\ \omega_t/(2\pi) &= 45.62\,\mathrm{kHz}, & \hbar\omega_t/k_{\mathrm{B}} &= 2.190\,\mu\mathrm{K}, & 2\pi/\omega_t &= 21.92\,\mu\mathrm{s}. \end{aligned} \tag{27}$$

Even though the lattice is deep in recoil units, each second-order axis has an rms mismatch phase of 0.7641 radians across a site. Averaging these phases gives $b_2 = e^{-(0.7641)^2/2}$; multiplying two independent axes gives $Q = 0.557734$. The phase spread $nK\sigma_x$, rather than depth alone, controls the overlap and explains the strong n^2 sensitivity. Ground-state occupation requires temperature small compared with $\hbar\omega_t/k_{\mathrm{B}}$. For example, Eq. (3) gives $Q = 0.48137$ at $T = 1\,\mu\mathrm{K}$, with the same depth and geometry.

The equilibrium density supplies an equal-time spatial average, including zero-point width. Replacing the atomic motion by this static profile additionally requires little motional evolution over the converter response time τ_{resp}: $\omega_t\tau_{\mathrm{resp}} \ll 1$, with motional sidebands unresolved by the relevant response. This is distinct from requiring small $nK\sigma_x$; the Gaussian phase average above does not expand in that quantity. The timescale check for the converter benchmark is given below. The model predicts the coherent response and does not calculate the inelastic motional spectrum.

B. Converter example.

Use the collective couplings and linewidths

$$\begin{aligned} \frac{(G, G_\mu, \Omega)}{2\pi} &= (15.8474565,\ 2.09170145,\ 5.52497339)\,\mathrm{MHz}, \\ \frac{(\kappa_o, \kappa_m, \Gamma, \gamma_r)}{2\pi} &= (10,\ 5,\ 6,\ 0.1)\,\mathrm{MHz}. \end{aligned} \tag{28}$$

All detunings are zero, and $\gamma_{m,\mathrm{add}} = \gamma_r$ reproduces the notebook convention. Consistently dividing every rate and frequency by 2π leaves the efficiency unchanged, allowing direct use of these MHz values. For unit port fractions and the finite-depth Q of Eq. (26):

Model	On-resonance efficiency
Matched profiles, $Q = 1$	82.0957%
Naive probability scaling, $Q^2\eta(Q = 1)$	25.5373%
Only $G_\mu \mapsto QG_\mu$; orthogonal pair omitted	51.5929%
Six modes with $Q = 0.557734$	**47.0824**%

Here $\Sigma_\perp(0)/(2\pi) = 0.147394\,\mathrm{MHz}$ and the additional full linewidth is $2\,\mathrm{Re}\,\Sigma_\perp(0)/(2\pi) = 0.294787\,\mathrm{MHz}$. Port fractions $\eta_o = \eta_m = 0.98$ give 45.21798% at unchanged total linewidths, before further path losses.

For this benchmark, $A/(2\pi) = 5.20468\,\mathrm{MHz}$ and $B/(2\pi) = 2.76394\,\mathrm{MHz}$ on resonance. The matching parameter falls from $\mathcal{C} = 0.61947$ at $Q = 1$ to 0.19270 at the finite-depth overlap. Both cases are below one, so reduced overlap worsens matching. At $T = 1\,\mu\mathrm{K}$, the smaller overlap quoted above would give $\eta(0) = 38.15\%$ with all converter parameters unchanged. Setting only $\gamma_{m,\mathrm{add}} = 0$ instead gives 48.54% at $T = 0$; this isolates the effect of the unassigned extra microwave-loss assumption.

The six-mode homogeneous drift matrix provides a response-time check. At the finite-depth benchmark its slowest amplitude decay rate is $2\pi \times 1.4533\,\mathrm{MHz}$, corresponding to $\tau_{\mathrm{resp}} = 0.1095\,\mu\mathrm{s}$. Thus $\omega_t\tau_{\mathrm{resp}} = 0.0314 \ll 1$, and the 45.62 kHz trap spacing is small compared with the response relaxation scale. This supports the frozen-motion approximation for this benchmark; parameters producing a much narrower resonance require a new check and possibly a dynamical motional model.

C. Where the unconverted input goes.

For the drive phase convention in Eq. (17), the optical reflected field is $a_{\rm out} = a_{\rm in} + i\sqrt{\kappa_{o,\rm ex}}\, a$. With incident flux $\Phi = |a_{\rm in}|^2$, define

$$\begin{aligned} \mathcal{R}_o &= \left|1 + i\sqrt{\kappa_{o,\rm ex}}\, a/a_{\rm in}\right|^2, \\ \mathcal{L}_{\rm at} &= \frac{\Gamma(|p|^2 + |p_\perp|^2) + \gamma_r(|r|^2 + |r_\perp|^2)}{\Phi}, \\ \mathcal{L}_o &= \frac{(\kappa_o - \kappa_{o,\rm ex})|a|^2}{\Phi}, \qquad \mathcal{L}_m = \frac{(\kappa_m - \kappa_{m,\rm ex} + \gamma_{m,\rm add})|b|^2}{\Phi}. \end{aligned} \tag{29}$$

The passive linear model satisfies $\mathcal{R}_o + \eta + \mathcal{L}_{\rm at} + \mathcal{L}_o + \mathcal{L}_m = 1$. For unit port fractions, $\mathcal{L}_o = 0$ and only $\gamma_{m,\rm add}$ contributes to $\mathcal{L}_m$. The benchmark budget, in percent of incident photon flux, is

Channel	$Q = 1$	$Q = 0.557734$
Microwave output	82.0957	47.0824
Optical reflection	3.4748	37.3236
Total atomic dissipation	12.7875	14.6524
Added microwave dissipation	1.6419	0.9416
Total (before rounding)	100	100

The orthogonal pair accounts for 2.7759% of incident flux, included in total atomic dissipation. The conversion falls by 35.0133 percentage points, while reflection rises by 33.8487 points and total atomic dissipation by only 1.8648 points. Most of the conversion reduction therefore appears as increased reflection. The orthogonal channel changes the coupled response as well as dissipating: its direct loss is not equal to the difference between the four- and six-mode efficiencies. This is why neither projection loss nor added linewidth alone explains the output.

D. Scope and checks.

The closure requires weak excitation, frozen positions, uniform control and atomic linewidths, and one effective optical profile. Anharmonic motion, resolved sidebands, differential light shifts, or further independent standing-wave channels require an extended susceptibility. Mathematica checks reproduce the six-equation solution and $Q = 1$ limit. Direct matrix solutions also verify the effective-cavity reduction, the numerical tables, and photon-flux conservation. These are checks of algebra and numerical implementation; they do not establish exact lattice matching or a microscopic origin for $\gamma_{m,\rm add}$. Numerical predictions remain conditional on the stated geometry and model assumptions.

E. Using the companion Mathematica notebook.

`Abhishek_Converter_QPM.nb` solves the six equations, eliminates the orthogonal pair, checks $Q = 1$, and evaluates the example. Its κ, κ_μ are κ_o, κ_m here; other physical symbols are unchanged. The calculator's $G_{\rm opt}, G_{\rm mm}$ are G, G_μ. Set the auxiliary parameter $\epsilon = 1$. Both notebooks specialize to $\gamma_{m,\rm add} = \gamma_r$; the independent $\gamma_{m,\rm add}$ in this note generalizes their microwave damping. All quoted notebook comparisons use that specialization.

The notebook's forcing parameter `Ein` remains in its amplitude `ConverterEfficQ` $= \sqrt{\kappa_o \kappa_m}\, b$. The input-normalized response, up to the port phase, is

$$S_{mo} = \sqrt{\eta_o \eta_m}\, \frac{\texttt{ConverterEfficQ}}{\texttt{Ein}}, \qquad \eta = \eta_o \eta_m\, \texttt{NormalizedEfficiencyQ}, \tag{30}$$

Here $\epsilon = 1$ and `NormalizedEfficiencyQ` $=$ `EfficiencyQ`$/$`Ein`^{2} for real `Ein`. Examples set `Ein`=1; normalized linear efficiency is independent of drive amplitude.

`SigmaPerp` implements Eq. (18); `OnResonanceEfficiencyQ` implements Eq. (24) with unit port fractions and $\gamma_{m,\rm add} = \gamma_r$.

`parmsScreenshot` gives Eq. (28), while `parmsOriginal` retains the original notebook's different parameters. Keep Q symbolic until applying `qRule`. The notebook's x, z labels denote the same two matched axes called x, y here.

Its `etaOpt`,`etaMM` are η_o, η_m, applied by `EfficiencyQWithPorts` at fixed linewidths. That function multiplies the unnormalized `EfficiencyQ`; use `Ein=1` or divide its result by `Ein`^{2} for real nonzero drive.

---

[1] I. Bloch, J. Dalibard and W. Zwerger, *Rev. Mod. Phys.* **80**, 885 (2008), Sec. II. doi:10.1103/RevModPhys.80.885.

[2] A. Kumar et al., *Nature* **615**, 614–619 (2023). doi:10.1038/s41586-023-05740-2.

[3] A. Kumar et al., *Quantum-limited millimeter wave to optical transduction*, arXiv:2207.10121 (2022), Supplement B.1–B.3. Full text and supplement.

[4] Baseline: `Abhishek_Converter.nb`; QPM implementation: `Abhishek_Converter_QPM.nb`. Code and calculator: AVKGitHub1/transduction_calcs, commit `30da8841b3c9`. The Q extension is derived here.